TECHNICAL REPORT: MODULE BASE SETTING SYSTEM

1. OVERVIEW AND CONCEPT

1.1. Problem Statement

In our current Gateway system, we need to support multiple types of wireless modules (BLE, Zigbee, LoRa, Thread) from various manufacturers. Each module has:

- Different command syntax in AT format, ASCII format, binary format (e.g., AT+NAME=xxx, AT+SETNAME=xxx, AT+DEVNAME=xxx)
- Different communication protocols (UART, SPI, I2C, USB)
- Different GPIO control mechanisms (Reset, Wake, Boot mode pins)
- Different timeout values and response formats

If we hardcode for each module, the code will be:

- Very long and complex
- Difficult to maintain and extend
- Requires firmware recompilation for every new module
- Inflexible when customers want to change modules

1.2. Module Base Setting Solution

Instead of standardizing COMMANDS (not feasible), we standardize by FUNCTION.

Example:

```
Function: MODULE_SET_NAME
- STM32WB55 BLE: AT+NAME=DeviceName\r\n
- HC-05 BLE: AT+NAME:DeviceName\r\n
- ESP32 BLE: AT+BLENAME=DeviceName\r\n
- nRF52 BLE: AT+GAPDEVNAME=DeviceName\r\n
```

All perform the SAME FUNCTION: Set device name.

1.3. Why Choose This Approach?

Reason 1: 70-80% of Functions are IDENTICAL

- Almost all BLE modules have: Reset, Set Name, Connect, Disconnect, Scan, Send Data
- Differences are only in: Command syntax, GPIO pins, timeout values

Reason 2: FLEXIBILITY

- No need to recompile firmware when adding new modules
- Simply create a new JSON file for that module
- PC App or Server sends JSON config to Gateway

Reason 3: EASY MAINTENANCE

- Firmware code is GENERIC, not module-specific
- JSON files are EXTERNAL CONFIG, easy to edit
- Testing and debugging are much easier

Reason 4: GOOD SCALABILITY

- Add new module: Just write a JSON file
- Add new function: Just update function_id enum and JSON file

- Support multiple stacks in parallel (Stack 0, Stack 1)

2. WHY COMPATIBLE WITH MULTIPLE MODULES?

2.1. Abstraction Principle

The Module Base Setting system is separated into 3 layers:

```
+---------------------+
|   JSON Config File  |  <-- MODULE SPECIFIC (External)
+---------------------+
          |
          v
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|  Function Handler   |  <-- GENERIC (Firmware)
|   (ble_handler.c)   |
+---------------------+
          |
          v
+---------------------+
|  BSP/Driver Layer   |  <-- HARDWARE SPECIFIC (Firmware)
|  (UART/SPI/I2C/USB) |
+---------------------+
```

2.2. Which Functions are STANDARD?

We analyzed 8 popular BLE modules and found:

Function	STM32WB55	HC-05	ESP32	nRF52	CC2640	DA14580	BL600	BGM1
HW Reset	GPIO	GPIO	GPIO	GPIO	GPIO	GPIO	GPIO	GPIO
SW Reset	AT+RESET	AT+RESET	AT+RST	AT+RESET	AT+RESET	AT+R	AT+ATZ	AT+RESET
Set Name	AT+NAME=	AT+NAME:	AT+BLENAME=	AT+GAPDEVNAME=	AT+NAME=	AT+REN	AT+CFG 532	gatt_devic
Connect	AT+CONNECT=	AT+LINK=	AT+BLECONN=	AT+GAPCONNECT=	AT+CON=	AT+CON	AT+BTCconnect	ble_gap_c
Scan	AT+SCAN	AT+INQ	AT+BLESCAN	AT+GAPSCAN	AT+DISC	AT+SCAN	AT+INQUIRY	ble_gap_d

Conclusion:

- **FUNCTIONS**: 100% IDENTICAL
- **COMMANDS**: 100% DIFFERENT

Therefore, if we standardize by FUNCTION, we can support ALL modules just by changing the JSON file.

2.3. Complete Abstraction Mechanism

2.3.1. Command Abstraction

JSON file contains:

- Function name: MODULE_SET_NAME
- Command template: AT+NAME={PARAM}\r\n

Firmware executes:

- Calls function with parameter "MyDevice"

- Firmware automatically replaces {PARAM} with "MyDevice"
- Sends: "AT+NAME=MyDevice\r\n"

2.3.2. Protocol Abstraction

JSON file contains:

- Port type: UART
- Parameters: baudrate 115200, parity none, stopbit 1

Firmware calls generic bus functions:

- module_bus_write() and module_bus_read()

System automatically routes to:

- module_uart_comm_write() if port_type = "uart"
- module_spi_comm_write() if port_type = "spi"
- module_i2c_comm_write() if port_type = "i2c"
- module_usb_comm_write() if port_type = "usb"

2.3.3. GPIO Abstraction

JSON file contains:

- Function: MODULE_HW_RESET
- GPIO start: RST pin = LOW
- Delay start: 100ms
- GPIO end: RST pin = HIGH
- Delay end: 1000ms

Firmware execution sequence:

1. Execute GPIO start sequences (pull RST LOW)
2. Wait delay_start_ms (100ms)
3. Send command (if any)
4. Wait for response
5. Execute GPIO end sequences (pull RST HIGH)
6. Wait delay_end_ms (1000ms)

2.4. Compatibility Conclusion

The Module Base Setting system can support ANY module that has:

- AT-based or Binary protocol commands
- UART/SPI/I2C/USB communication
- GPIO control for Reset/Boot/Wake
- Timeout and response validation

Only need to create 1 JSON file describing that module.

3. ROLE OF PC APP / SERVER

3.1. PC App: Gateway Configuration Tool

PC App (Gateway_Config_Tool_v4) is responsible for:

3.1.1. Module Selection Interface

Provides user interface for:

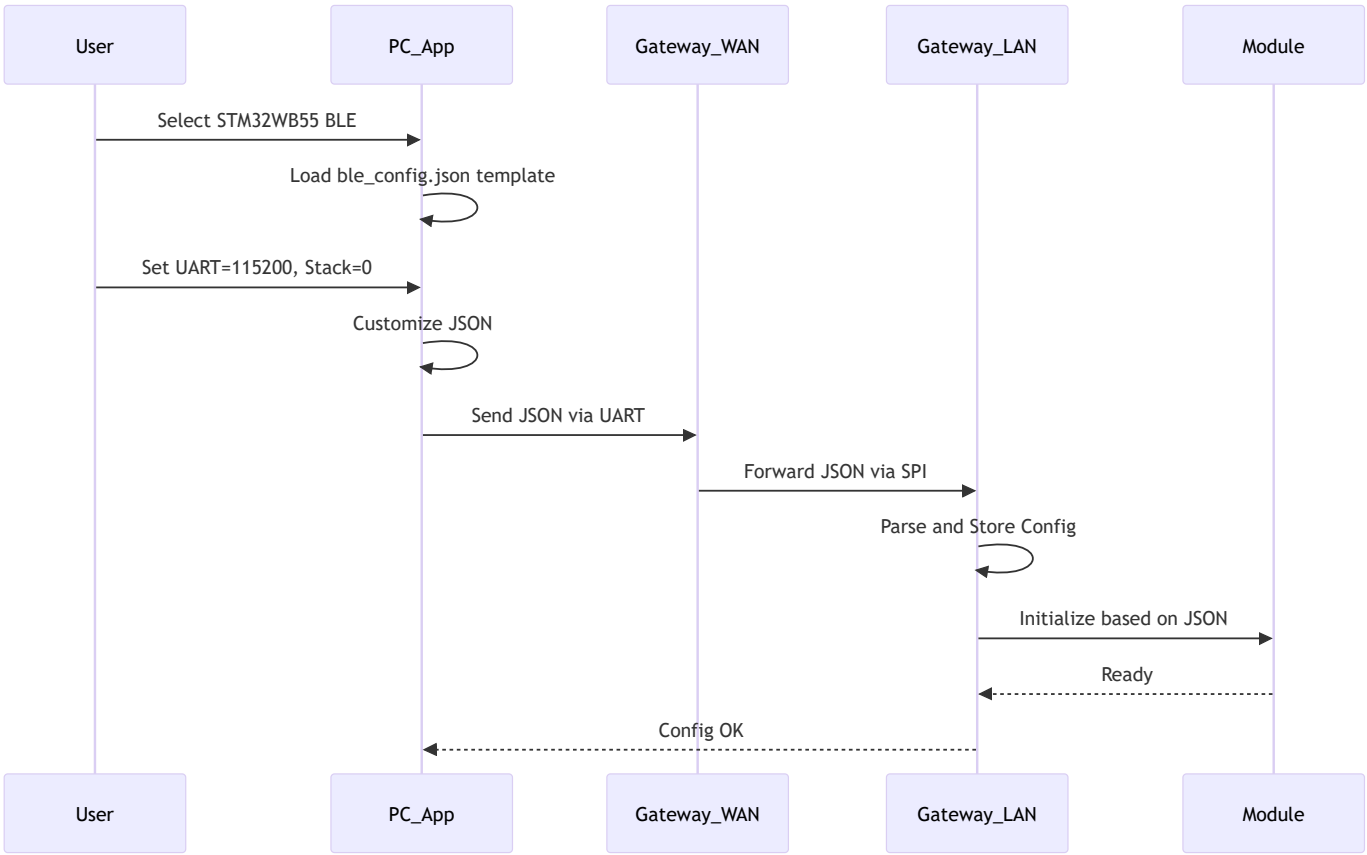
- Selecting module type (STM32WB55 BLE, nRF52840, HC-05, etc.)
- Configuring communication parameters (UART baudrate, SPI speed, etc.)
- Loading custom JSON configurations
- Generating and sending config to Gateway

3.1.2. JSON Config Generation

When user selects a module, the App will:

1. Load template JSON from database (embedded or cloud)
2. Customize based on user input (baudrate, GPIO mapping)
3. Validate JSON format
4. Send JSON string to Gateway via UART

Data Flow:



3.1.3. Module Testing Interface

After configuration, PC App allows direct testing:

- Hardware Reset, Software Reset, Get Info buttons
- Set Name input and Apply
- Scan Devices function
- Connect to specific device
- Response log display showing AT commands and responses

3.2. Server: Remote Configuration Management

Server (MQTT/HTTP/CoAP) is responsible for:

3.2.1. Centralized Config Storage

Cloud server maintains configuration repository:

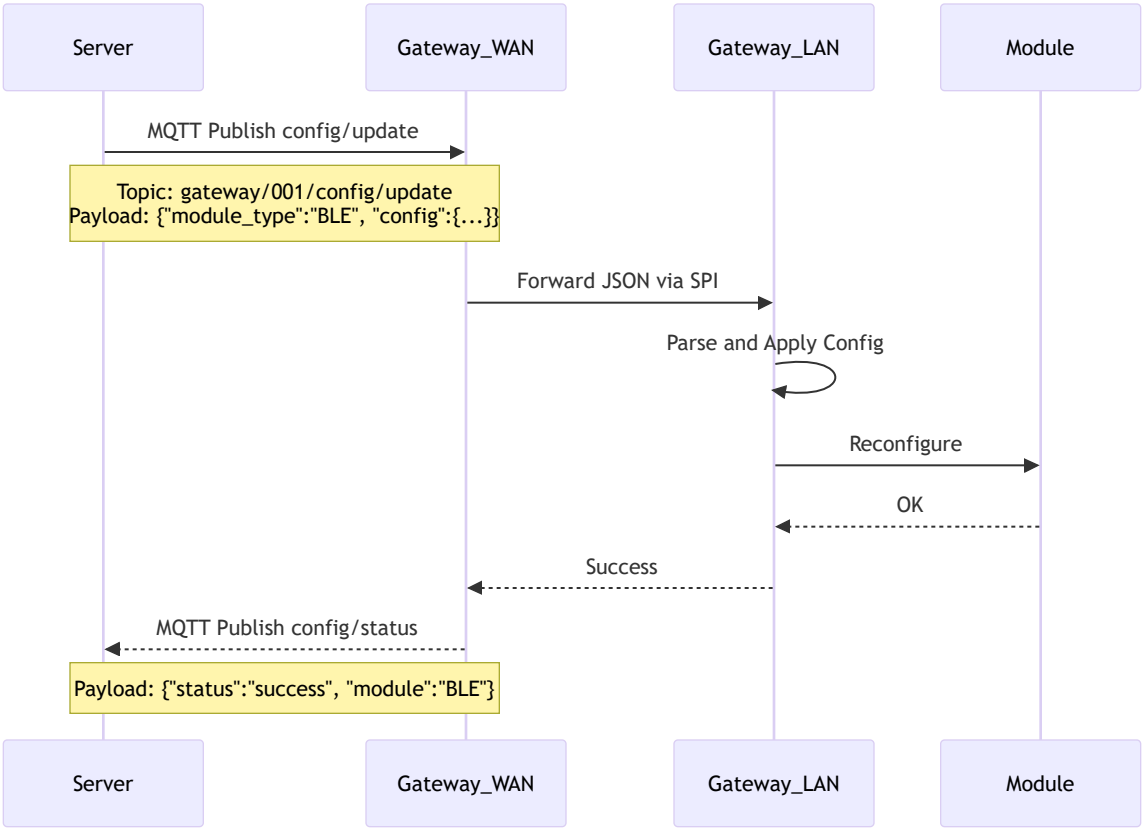
- devices/gateway_001/ble_config.json
- devices/gateway_001/zigbee_config.json
- devices/gateway_002/ble_config.json
- etc.

3.2.2. Remote Update Flow

MQTT Topic Structure:

- gateway/{device_id}/config/update (Server publishes JSON config)
- gateway/{device_id}/config/status (Gateway reports status)
- gateway/{device_id}/config/request (Gateway requests config)

Communication Flow:



3.2.3. Bulk Update Management

Server can push updates simultaneously to multiple gateways:

- Load new configuration (e.g., stm32wb55_v2.json)
- Iterate through all registered gateways
- Publish update to each gateway's MQTT topic
- Wait for confirmation with timeout
- Track success/failure for each gateway

3.3. Advantages of This Approach

1. **Zero Downtime Update**: Configuration updates without Gateway restart
2. **Version Control**: Server stores history of all config versions
3. **Rollback**: Can revert to previous config if errors occur
4. **A/B Testing**: Test new config on subset of gateways before wide deployment
5. **Monitoring**: Server tracks configuration status of all gateways

4. ROLE OF BASEBOARD GATEWAY

4.1. Gateway Architecture

Gateway consists of 2 MCUs:

```
+---------------------------+        +---------------------------+
|      WAN MCU (ESP32)      |        |      LAN MCU (ESP32)      |
|---------------------------|        |---------------------------|
| - MQTT/HTTP/CoAP Client   |        | - Module Management       |
| - WiFi/4G Connection      | <-SPI->| - JSON Config Parser      |
| - Data Routing            |        | - UART/SPI/I2C/USB Driver |
| - OTA Update              |        | - Multi-Stack Support     |
+---------------------------+        +---------------------------+
         ^                                     ^
         |                                     |
         v                                     v
+------------------+             +---------------------------+
|   PC App /       |             |    Wireless Modules       |
|   Server         |             | - BLE (Stack 0)           |
+------------------+             | - Zigbee (Stack 1)        |
                                 | - LoRa, Thread, etc.      |
                                 +---------------------------+
```

4.2. WAN MCU: Internet Communication Handler

Main Responsibilities:

- Receive JSON config from PC App (via UART) or Server (via MQTT/HTTP)
- Forward JSON to LAN MCU via SPI
- Route sensor data from LAN MCU to Server
- Route control commands from Server to LAN MCU

Does NOT:

- Parse JSON (to reduce processing load)
- Know details about BLE/Zigbee/LoRa modules
- Acts only as a TRANSPARENT BRIDGE

4.3. LAN MCU: Module Configuration Controller

Main Responsibilities:

4.3.1. JSON Config Reception and Parsing

When receiving CONFIG_UPDATE_BLE_JSON command:

- Parse JSON string
- Store configuration in BLE handler
- Initialize communication port (UART/SPI/I2C/USB)
- Module is now ready to use

4.3.2. Function Execution Engine

When receiving command from Server or PC App:

Execution Steps:

1. Find function config from stored JSON
2. Build final command (replace {PARAM} placeholder)
3. Execute GPIO start sequences (if any) and wait delay_start
4. Send command via configured bus

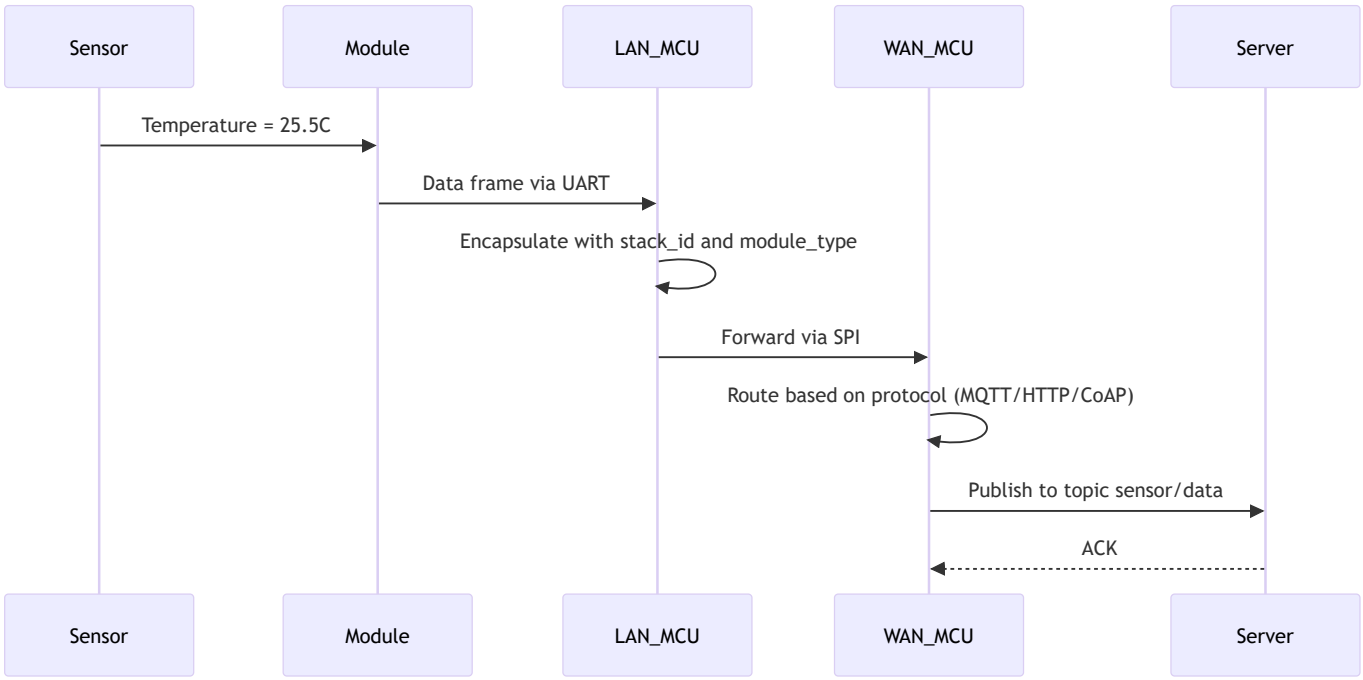
5. Wait and validate response with timeout
6. Execute GPIO end sequences and wait delay_end
7. Return result

4.3.3. Multi-Stack Management

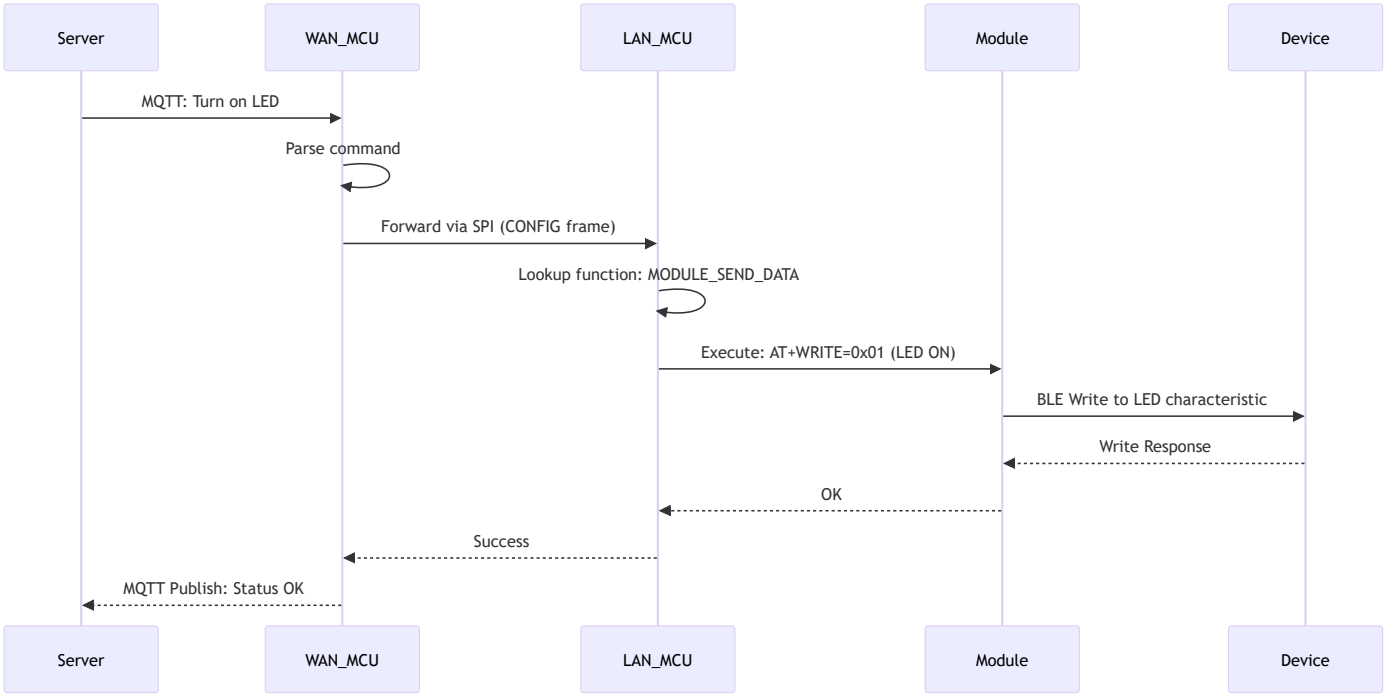
LAN MCU supports multiple modules running simultaneously:

- Stack 0: BLE Module (UART, 115200 baud)
- Stack 1: Zigbee Module (SPI, 2MHz)
- Both stacks run independently with separate tasks and queues

4.4. Data Flow: Sensor Data Upload



4.5. Data Flow: Control Command from Server



4.6. Critical Role of Gateway

Gateway is NOT just a simple bridge, it is:

1. **Configuration Manager**: Receives, parses, and applies module configuration
2. **Protocol Converter**: Converts between Internet protocols (MQTT/HTTP) and Module protocols (AT commands)
3. **Multi-Stack Coordinator**: Manages multiple modules simultaneously without conflicts
4. **Data Router**: Routes data between sensors and server
5. **Transparent Bridge**: PC App can connect directly to module through Gateway for testing

5. ROLE OF JSON FILE

5.1. Why JSON is Needed?

5.1.1. Human-Readable and Machine-Parsable

Compared to binary config or hardcoding:

- **BAD (Hardcode)**: Difficult to read, difficult to modify, requires firmware rebuild
- **GOOD (JSON)**: Easy to read, easy to modify, no rebuild needed

JSON provides structured configuration that both humans and machines can understand.

5.1.2. External Configuration

JSON is a separate file, not embedded in firmware:

- **Developer**: Edit JSON to test new modules
- **QA Team**: Create JSON test cases for automation
- **Customer**: Receive JSON from module vendor and import into system
- **Server**: Store and distribute JSON configs

5.1.3. Version Control Friendly

JSON file is easy to version control with Git:

- Clear diff showing what changed (command syntax, timeout values, etc.)
- Easy rollback to previous versions
- Branch and merge support for testing variants

5.1.4. Validation and Schema

JSON can be validated before applying:

- Define schema with required fields and data types
- Validate module_type is one of allowed values (BLE, Zigbee, LoRa, Thread)
- Ensure functions array contains valid entries
- Check timeout values are non-negative numbers
- Raise errors if validation fails

5.2. Complete JSON Structure

JSON file contains:

- Module identification (module_id, module_type, module_name)
- Communication configuration (port_type, parameters)
- Function definitions array (20 functions for BLE)

Each function includes:

- Function name

- Command template
- GPIO control sequences
- Timing parameters
- Expected response

5.3. JSON vs Alternatives

Method	Advantages	Disadvantages
Hardcode	Fast execution	Hard to modify, requires rebuild
Binary Config	Compact size	Hard to read, hard to debug
XML	Structured	Verbose, heavy parsing
INI File	Simple	No nested structure support
JSON	Human-readable, structured, lightweight parser	Slightly verbose

Conclusion: JSON is the best choice for this system.

6. ROLE OF EACH JSON COMPONENT

6.1. Module Metadata

Components:

- module_id: "002" (Unique identifier for each module in database)
- module_type: "BLE" (Module category - BLE, Zigbee, LoRa, Thread)
- module_name: "STM32WB_BLE_Gateway" (Display name for user interface)

Purpose:

- module_id: Unique identification for managing multiple modules
- module_type: Routes to correct handler (BLE handler, Zigbee handler, etc.)
- module_name: User-friendly name displayed in UI

Usage in Firmware:
Router checks module_type to determine which handler to use (ble_handler_set_config, zigbee_handler_set_config, etc.)

6.2. Communication Configuration

Structure:

- port_type: "uart", "spi", "i2c", or "usb"
- parameters: Specific to each port type

6.2.1. Port Type Selection

Based on port_type, firmware initializes the correct BSP driver:

- "uart" -> module_config_controller_init_uart()
- "spi" -> module_config_controller_init_spi()
- "i2c" -> module_config_controller_init_i2c()
- "usb" -> module_config_controller_init_usb()

6.2.2. Protocol Parameters

UART Parameters:

- baudrate: 9600, 115200, 921600, etc.
- parity: none, even, odd

- stopbit: 1, 2
- data_bits: 8 (default)

SPI Parameters:

- clock_speed: 1000000 (1MHz), 2000000 (2MHz)
- mode: 0, 1, 2, 3 (CPOL, CPHA)
- bit_order: MSB_FIRST, LSB_FIRST

I2C Parameters:

- address: 0x50, 0xA0, etc. (7-bit or 10-bit)
- clock_speed: 100000 (Standard), 400000 (Fast), 1000000 (Fast+)

USB Parameters:

- vid: Vendor ID
- pid: Product ID
- interface: Interface number

Why This Section is Important:

- Different modules use different protocols
- Same protocol but different parameters
- Firmware needs exact configuration to initialize driver properly

6.3. Function Name (Standardized Functions)

Purpose: Maps JSON entry to hardcoded function ID in firmware.

Hardcoded Function IDs:
Firmware defines 20 function IDs for BLE:

- BLE_FUNC_HW_RESET = 0
- BLE_FUNC_SW_RESET = 1
- BLE_FUNC_FACTORY_RESET = 2
- BLE_FUNC_GET_INFO = 3
- BLE_FUNC_SET_NAME = 4
- ... up to BLE_FUNC_ENTER_BOOTLOADER = 19

Lookup Process:
Parser maintains array of function names and performs string comparison to find matching ID.

Why Hardcode Function Names?

1. **Contract**: Ensures firmware and JSON config "speak the same language"
2. **Type Safety**: Compile-time checking, prevents typos
3. **Performance**: Fast lookup by integer instead of string comparison every time
4. **Documentation**: Function enum serves as natural documentation for developers

Module-Specific Functions:
Different module types can define their own functions:

- Zigbee: FORM_NETWORK, JOIN_NETWORK, PERMIT_JOIN, BIND_DEVICE, etc.
- LoRa: JOIN_OTAA, JOIN_ABP, SEND_UPLINK, SET_DATARATE, etc.

6.4. Command (Module-Specific Command Syntax)

Purpose:

- Contains actual command sent to module
- Supports placeholder {PARAM} for dynamic parameter passing

Parameter Substitution:
Firmware replaces {PARAM} placeholder with actual parameter value:

- Input: command = "AT+NAME={PARAM}\r\n", param = "MyDevice"
- Output: "AT+NAME=MyDevice\r\n"

Special Case - Empty Command:
For GPIO-only functions:

- command field is empty
- Firmware only executes GPIO sequences
- No command sent via UART/SPI

Example: MODULE_HW_RESET uses only GPIO control

Why This Field is Needed?

- **Flexibility**: Each module has its own command syntax
- **Dynamic Parameters**: No need to hardcode parameters in firmware
- **Testability**: Can test commands directly from PC App

6.5. GPIO Control

6.5.1. GPIO Start Control (Before Sending Command)

Structure:

- gpio_start_control: Array of GPIO actions
- Each action specifies pin name and state (HIGH/LOW)
- delay_start: Milliseconds to wait after GPIO sequences

Purpose:

- Control GPIO pins BEFORE sending command
- Example: Pull RST pin LOW to reset module

Execution Sequence:

1. Execute all GPIO start sequences (set pins to specified states)
2. Wait delay_start_ms
3. Proceed with command sending

Use Cases:

- **Hardware Reset**: Pull RST pin LOW -> wait 100ms -> Pull HIGH
- **Boot Mode**: Pull BOOT pin HIGH -> Reset -> Module enters bootloader
- **Wake from Sleep**: Toggle WAKE pin
- **Enable Module**: Pull EN pin HIGH

6.5.2. GPIO End Control (After Receiving Response)

Structure:

- gpio_end_control: Array of GPIO actions after command completes
- delay_end: Milliseconds to wait after GPIO end sequences

Purpose:

- Control GPIO pins AFTER command completes
- Example: Pull RST pin HIGH so module can run again

Execution Sequence:

1. After command sent and response validated
2. Execute all GPIO end sequences

3. Wait delay_end_ms
4. Function returns

Why GPIO Control is Important?

1. **Hardware Reset**: Some modules can only be reset via GPIO (no AT command)
2. **Boot Mode Selection**: Need to pull GPIO for module to enter bootloader mode
3. **Power Management**: Turn off module via GPIO to save power
4. **Multi-Module Control**: Each module has different pins, JSON maps correct pins

GPIO Mapping:
Hardware design defines physical pins:

- Stack 0 (BLE): RST=GPIO0, BOOT=GPIO1, WAKE=GPIO2
- Stack 1 (Zigbee): RST=GPIO3, BOOT=GPIO4, WAKE=GPIO5

JSON uses symbolic names ("RST", "BOOT", "WAKE")
Firmware resolves symbolic names to physical GPIO numbers based on stack_id

6.6. Expected Response (Validation)

Purpose:

- Validate that module's response is correct
- Set timeout for waiting

Response Validation Process:

1. Wait for response with specified timeout
2. Check if response contains expected string
3. If match found: response valid, function succeeds
4. If no match: response invalid, function fails

Why Validation is Needed?

1. **Reliability**: Ensure module actually received and processed command
2. **Error Detection**: Detect early if module has error or is busy
3. **Flow Control**: Know when module is ready for next command

Special Cases:

Empty Response (GPIO-only functions):

- expect_response = ""
- timeout = 0
- Firmware skips module_bus_read, returns immediately

Example: MODULE_WAKEUP (GPIO-only, no response expected)

Async Response (Scan functions):

- expect_response = "+SCAN:"
- timeout = 10000 (10 seconds)
- Module returns multiple lines during timeout period
- Firmware must handle streaming response (described in Section 8.3)

6.7. Timeout (Response Wait Time)

Purpose:

- Set maximum time to wait for module response (in milliseconds)
- Prevents firmware from blocking indefinitely

Common Timeout Values:

Function	Timeout	Reason
MODULE_SET_NAME	500ms	Simple command
MODULE_CONNECT	5000ms	Needs time for handshake
MODULE_START_DISCOVERY	10000ms	BLE scan takes time
MODULE_FACTORY_RESET	3000ms	Module needs to erase data and reset
MODULE_HW_RESET	0ms	GPIO-only, no response wait

Implementation:
Firmware tracks elapsed time from start and returns ESP_ERR_TIMEOUT if timeout expires before receiving response.

Why Timeout is Important?

1. **Prevent Deadlock**: Don't block if module doesn't respond
2. **User Experience**: PC App or Server knows immediately when error occurs
3. **Resource Management**: Free task to process other requests

6.8. Delay (Wait Time Between Steps)

Purpose:

6.8.1. delay_start (Before Sending Command)

- Wait for module to stabilize after GPIO control
- Example: After pulling RST LOW, wait 100ms before pulling HIGH

6.8.2. delay_end (After Receiving Response)

- Wait for module to complete processing
- Example: After reset, wait 1000ms for module to boot up

Timing Example:

Function execution timeline for MODULE_HW_RESET:

- t=0ms: Execute GPIO start (Pull RST LOW)
- t=0ms: Start delay_start timer
- t=100ms: delay_start expires
- t=100ms: Send command "AT+RESET\r\n"
- t=100ms: Start waiting for response (timeout=2000ms)
- t=150ms: Receive "OK"
- t=150ms: Validate response
- t=150ms: Execute GPIO end (Pull RST HIGH)
- t=150ms: Start delay_end timer
- t=1150ms: delay_end expires
- t=1150ms: Function returns ESP_OK
- **Total execution time: 1150ms**

Why Delay is Needed?

1. **Hardware Timing**: Module needs time to recognize GPIO signals
2. **Boot Time**: Module needs time to boot after reset
3. **Stabilization**: Module needs time for stable connection
4. **Specification Compliance**: Module datasheet may require specific delays

Example from Datasheet:
STM32WB55 Reset Timing:

- Assert RESET (LOW) for minimum 100us
- Deassert RESET (HIGH)

- Wait minimum 1ms before sending first command
- Boot time: typically 800ms, max 1000ms

Corresponding JSON config:

- delay_start: 100ms (safe margin)
- delay_end: 1000ms (cover max boot time)

7. IMPLEMENTATION APPROACH

7.1. Layered Architecture

```
+-----------------------------------------------------------+
|                    APPLICATION LAYER                      |
|  (ble_handler_task.c, zigbee_handler_task.c)             |
|  - Receive commands from uplink/downlink queues          |
|  - Call middleware functions                              |
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+-----------------------------------------------------------+
|                    MIDDLEWARE LAYER                       |
|  (ble_handler.c, json_ble_config_parser.c,               |
|   module_config_controller.c)                             |
|  - Store JSON config in RAM                               |
|  - Execute functions based on config                      |
|  - Abstract away BSP details                              |
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|                    BSP LAYER                              |
|  (module_uart_comm.c, module_spi_comm.c, etc.)           |
|  - Hardware-specific drivers                              |
|  - UART/SPI/I2C/USB read/write functions                 |
|  - GPIO control                                           |
+-----------------------------------------------------------+
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+-----------------------------------------------------------+
|                    HARDWARE LAYER                         |
|  (ESP32 peripherals, STM32WB55 module)                   |
+-----------------------------------------------------------+
```

7.2. Data Structures

Key Components:

JSON Module Config (Parsed Storage):

- Stores function configurations after parsing
- Includes function_id, command, GPIO sequences, timeouts
- Array of 20 function configs for BLE

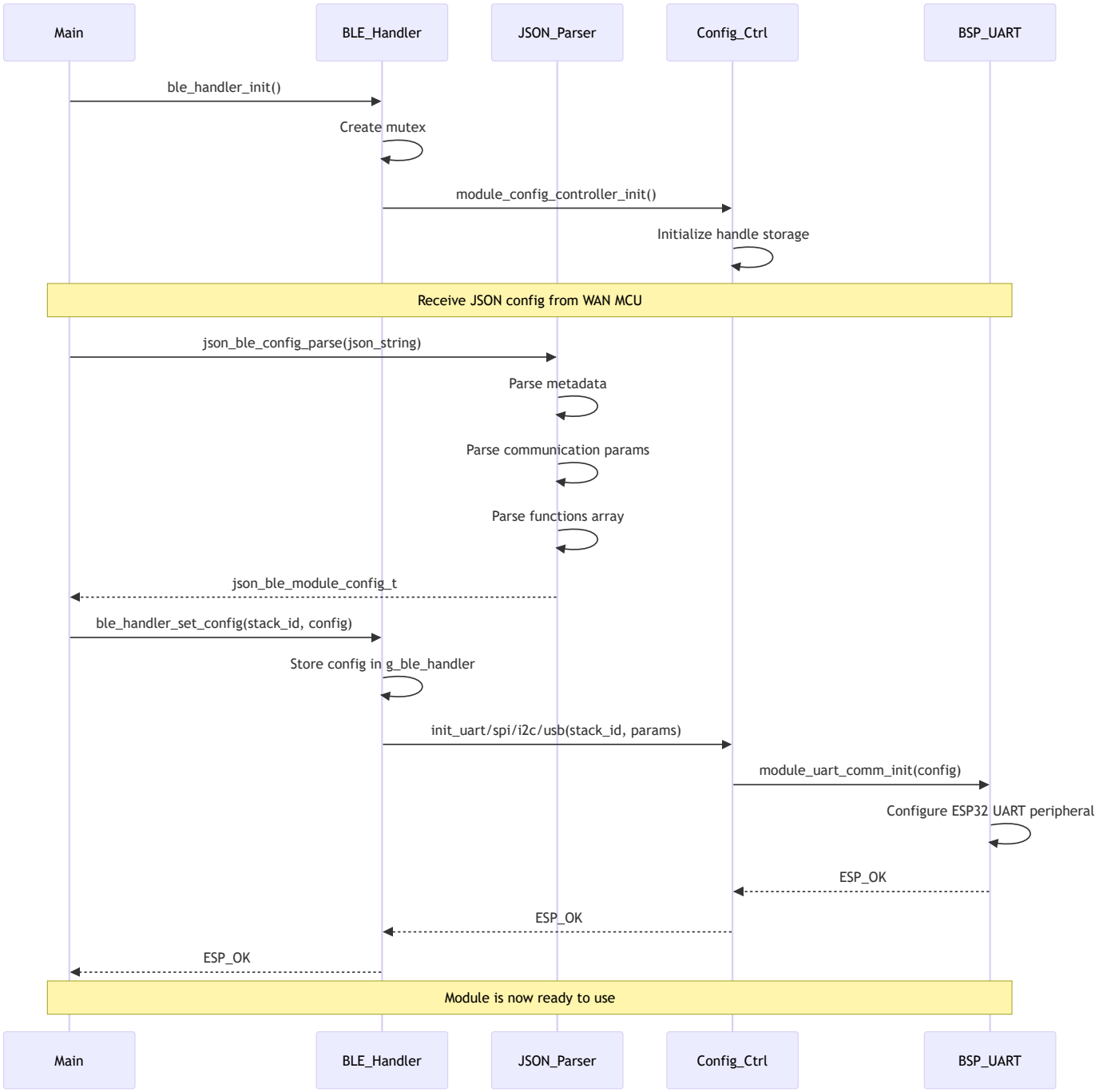
BLE Handler Config (Runtime Storage):

- Runtime storage for quick access
- Contains GPIO pin numbers and states
- Timing parameters (delays, timeouts)

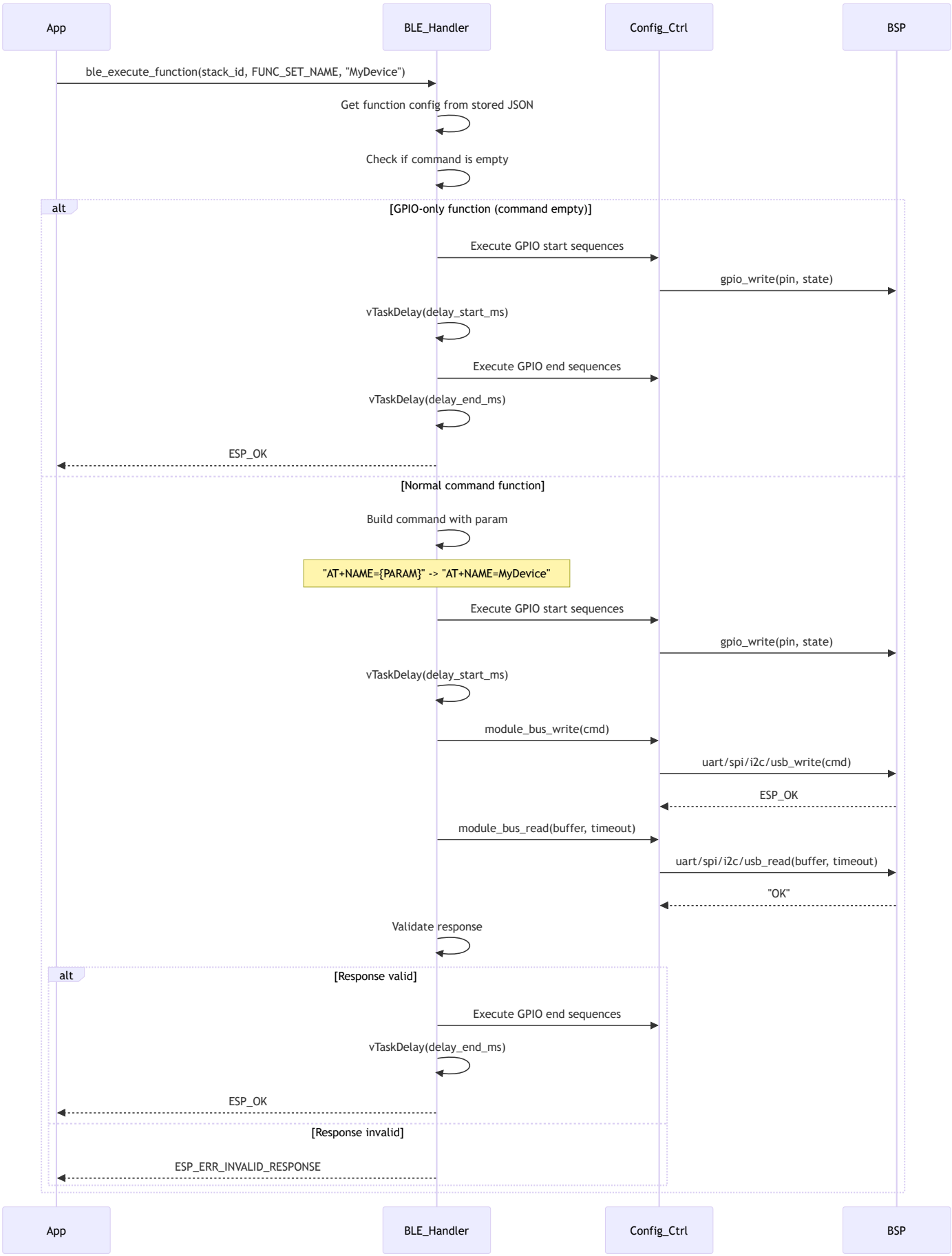
Global Handler State:

- Maintains state for both stacks (Stack 0 and Stack 1)
- Stores configuration for each stack
- Tracks connected devices per stack
- Uses mutex for thread-safe access

7.3. Initialization Flow



7.4. Function Execution Flow



7.5. Key Implementation Details

7.5.1. Config Storage Strategy

Problem: JSON string is large (2-5KB), parsing every time is slow.

Solution: Parse once, store config in RAM.

Process:

- Parse JSON into structured config
- Store entire config in global handler structure
- Access via O(1) lookup by function_id
- Memory overhead: ~2KB RAM per stack

7.5.2. Function Lookup Optimization

Optimization: O(1) lookup by function_id (integer index)

- Get function config directly from array
- Check if function is available in JSON config
- Return pointer to function config
- No need to loop through all functions

Performance: Instant lookup, no iteration needed.

7.5.3. Multi-Stack Isolation

Each stack has independent:

- Config storage
- UART/SPI/I2C/USB handles
- Device tracking
- Task queues

Example: Two BLE modules running simultaneously

- Stack 0: Scanning for devices
- Stack 1: Connecting to specific device
- No conflict, fully independent operation

7.5.4. Error Recovery

Retry Mechanism:

- Define max retry count (e.g., 3 retries)
- On timeout error, wait 500ms and retry
- Other errors don't trigger retry
- Log retry attempts for debugging

7.6. Memory Usage Analysis

Per Stack:

- Config storage: ~2KB
- Device tracking: ~1KB (16 devices x 64 bytes)
- Task stack: 4KB (uplink) + 4KB (downlink)
- Queues: ~2KB (20 items x 100 bytes)

Total for 2 stacks: ~26KB

Conclusion: Acceptable for ESP32 (520KB RAM).

8. BLE MODULE IMPLEMENTATION

8.1. BLE Function Set (20 functions)

8.1.1. Core Functions (15 mandatory)

0. MODULE_HW_RESET

- GPIO-only function
- Pull RST pin LOW -> wait 100ms -> Pull HIGH -> wait 1s
- Module reboots

1. MODULE_SW_RESET

- Software reset via command: AT+RESET
- Module will reboot and take 1-2s

2. MODULE_FACTORY_RESET

- Erase all config, restore to defaults: AT+RESTORE
- Takes longer than SW_RESET

3. MODULE_GET_INFO

- Get information: version, MAC address, firmware build
- Command: AT+VER
- Response: "+VER:1.2.3,MAC:AA:BB:CC:DD:EE:FF"

4. MODULE_SET_NAME

- Set BLE device name (advertising name)
- Command: AT+NAME={PARAM}
- Usage: Set device name to "SensorNode1"

5. MODULE_SET_COMM_CONFIG

- Configure module UART (baudrate, parity)
- Command: AT+UART={PARAM}
- Usage: Set to 921600 baud

6. MODULE_SET_RF_PARAMS

- Configure TX power, RF channel
- Command: AT+RF={PARAM}

7. MODULE_ENTER_CMD_MODE

- Enter AT command mode (if in transparent mode)
- Command: AT+CMDMODE

8. MODULE_ENTER_DATA_MODE

- Enter transparent data mode (bypass AT commands)
- Command: AT+DATAMODE={PARAM}

9. MODULE_START_BROADCAST

- Start advertising (Peripheral mode)
- Command: AT+ADV=1
- Central devices can scan and connect

10. MODULE_CONNECT

- Connect to remote BLE device (Central mode)
- Command: AT+CONNECT={PARAM}

- Usage: Connect to device with MAC address

11. MODULE_DISCONNECT

- Disconnect from device
- Command: AT+DISCONNECT={PARAM}

12. MODULE_GET_CONNECTION_STATUS

- List currently connected devices
- Command: AT+LIST
- Response: "+LIST:1,AA:BB:CC:DD:EE:FF,-65"

13. MODULE_ENTER_SLEEP

- Enter low-power mode
- Command: AT+SLEEP

14. MODULE_WAKEUP

- GPIO-only function
- Toggle WAKE pin or send dummy byte to wake module

8.1.2. Optional Functions (5 for advanced features)

15. MODULE_START_DISCOVERY

- Scan BLE devices nearby
- Command: AT+SCAN
- Response: multiple lines during 10s timeout
- Format: +SCAN:MAC,-RSSI,Name

16. MODULE_SEND_DATA

- Send data to connected device
- Command: AT+WRITE={PARAM}
- Usage: Send hex data

17. MODULE_GET_DIAGNOSTICS

- Get RSSI, link quality, connection parameters
- Command: AT+INFO={PARAM}

18. MODULE_SET_SECURITY_CONFIG

- Configure security: pairing, bonding, encryption
- Command: AT+SEC={PARAM}

19. MODULE_ENTER_BOOTLOADER

- Enter bootloader mode for OTA firmware update
- GPIO sequence: BOOT=HIGH, RST=LOW, then RST=HIGH

8.2. BLE Task Architecture

```
+-------------------+       +-------------------+
|  BLE Uplink Task  |       | BLE Downlink Task |
|    (Stack 0)      |       |    (Stack 0)      |
+-------------------+       +-------------------+
         |                           |
         v                           v
+-----------------------------------------+
|         BLE Handler Middleware            |
|  (ble_handler_execute_function)          |
+-----------------------------------------+
         |                           |
         v                           v
+-------------------+       +-------------------+
|   Module Config   |       |   BSP Drivers     |
|    Controller     |       | (UART/SPI/I2C/USB)|
+-------------------+       +-------------------+
```

8.2.1. Uplink Task (Module -> Server)

Function: Receive sensor data from module, forward to WAN MCU.

Process:

- Wait for data from module (via UART interrupt, ISR pushes to queue)
- Encapsulate with module info (stack_id, module_type)
- Forward to WAN MCU via SPI
- Track failed transmissions

8.2.2. Downlink Task (Server/App -> Module)

Function: Receive control commands from WAN MCU, execute functions.

Process:

- Wait for command from WAN MCU via queue
- Parse frame type (CONFIG_UPDATE, FUNCTION_EXEC, SCAN_REQUEST)
- Execute corresponding BLE function
- Send response back to WAN MCU

Frame Types:

- FRAME_TYPE_CONFIG_UPDATE: Apply JSON config
- FRAME_TYPE_FUNCTION_EXEC: Execute specific function
- FRAME_TYPE_SCAN_REQUEST: Start discovery and stream results

8.3. BLE-specific Features

8.3.1. Device Tracking

Purpose: Track connected devices per stack

- Store MAC address, connection time, last activity, RSSI
- Maximum 16 devices per stack
- Support add, update, and remove operations

Use Cases:

- Monitor connection status
- Detect disconnected devices
- Report device list to server

8.3.2. Discovery Result Caching

Purpose: Cache last scan results

- Store up to 20 discovered devices
- Include MAC address, RSSI, device name
- Allow re-query without scanning again

Why Caching?

- BLE scan takes time (5-10s)
- PC App or Server can query recent scan results
- Avoid continuous scanning

8.3.3. Streaming Response Handler

Problem: Scan function returns multiple lines of response over 10s.

Solution: Stream each line to WAN MCU immediately upon receipt.

Process:

1. Send scan command
2. Stream responses for specified duration
3. Parse each scan result line
4. Forward immediately to WAN MCU
5. Also cache locally for later queries

Format: +SCAN:AA:BB:CC:DD:EE:FF,-65,DeviceName

END OF REPORT